

Server Setup Guide: What IT Needs to Download and Install

Target machine: 192.168.1.121 , the Web-Team server.

Purpose: this server will host the Fute Portal's backend (the code that handles logins, tickets, approvals etc.) and its database, entirely on the office network. Once this is set up, the company will not need to pay for any cloud hosting service (like Firebase or Vercel) to run this application. Everything will run locally, on this one machine.

This guide is written so a non-developer on the IT team can follow it step by step. It only covers **installing the tools**. The actual application code will be copied onto this machine separately by the development team, after these tools are in place.

1. What software needs to be downloaded

Three pieces of software need to be installed on this server. Two are required, one is optional.

Order	Software	What it's for	Download link
1	Node.js (LTS version)	This is the runtime that lets the backend code (written in JavaScript) actually run on the server. Without this, none of the application's code will work.	https://nodejs.org/en/download
2	MongoDB Community Server	This is the database. It's where all the application's data will actually be stored: employee records, tickets, complaints, approvals, everything. This is the "storage" part of the whole setup.	https://www.mongodb.com/try/download/community
3	Google Chrome (optional)	Only needed if this server doesn't already have a browser installed. Useful for opening the app locally to check it's working after setup.	https://www.google.com/chrome/

Note: You do not need to separately download anything else. Two small tools called PM2 and serve are needed later, but those install automatically through a single command once Node.js is already installed. No separate download page for those.

2. Step-by-step installation

Step 1: Install Node.js

1. Go to <https://nodejs.org/en/download> and download the Windows Installer (the file ending in `.msi`). Pick the version labeled **LTS** (this means "Long Term Support" and is the stable, recommended version).

2. Double-click the downloaded file to run it.
3. Click "Next" through the installer and accept all the default options. There's nothing to change.
4. Click "Install", then "Finish" when it's done.

Step 2: Install MongoDB (the database)

1. Go to <https://www.mongodb.com/try/download/community>
2. Under "Version", leave it as the current version. Under "Platform", choose **Windows**. Under "Package", choose **msi**.
3. Click download, then run the downloaded `.msi` file.
4. When the setup wizard asks for a setup type, choose **"Complete"** (not "Custom").
5. When it asks about "Install MongoDB as a Service", leave that box **checked**. This means MongoDB will start automatically every time the server restarts, without anyone needing to manually start it.
6. You can leave "Install MongoDB Compass" checked too if you'd like a visual tool to browse the database later, though it isn't required.
7. Click "Install", then "Finish".

Step 3: Confirm both installed correctly

Open PowerShell (search for it in the Start menu, right-click, and choose "Run as Administrator"). Then type each of the following commands one at a time and press Enter:

```
node -v
```

This should print something like `v20.x.x`. If it prints a version number, Node.js is installed correctly.

```
mongosh --version
```

This should also print a version number. This confirms MongoDB's command-line tool is installed and working.

If either command says "not recognized", it usually means the computer needs to be restarted once after installation so the new software is recognized by the system.

Step 4: Turn on "replica set" mode for MongoDB

This step is required because the application uses a database feature (called a "transaction") that only works when MongoDB is running in this mode. It sounds technical, but it's just three small actions:

1. Open PowerShell and run this command to find the exact folder MongoDB installed itself into (do not type a version number yourself, this command finds it automatically):

```
notepad (Get-ChildItem "C:\Program Files\MongoDB\Server\*\bin\mongod.cfg").FullName
```

This opens `mongod.cfg` directly in Notepad without you needing to know or guess the version number.

2. If that command gives an error, it means MongoDB was installed to a different, custom folder. In that case, use Windows Search (the magnifying glass next to the Start button) and search for `mongod.cfg` to find where it actually is, then open that file with Notepad.
3. Scroll to the bottom of the file and add these two lines exactly as shown:

```
replication:  
replSetName: "rs0"
```

4. Save the file and close Notepad.
5. Restart the MongoDB service so it picks up this change:
 - Press `Windows key + R`, type `services.msc`, and press Enter.
 - Find "MongoDB Server" in the list, right-click it, and choose "Restart".
6. Open PowerShell again and type:

```
mongosh
```

This opens the MongoDB shell. Inside it, type this one command and press Enter:

```
rs.initiate()
```

This only needs to be run **once, ever**, and finishes the replica set setup. Type `exit` afterward to close the shell.

Step 5: Install two small helper tools

These two tools help keep the application running smoothly. They are not separate programs to download; they install directly through Node.js's own installer command.

In PowerShell, type:

```
npm install -g pm2 serve
```

This may take a minute. When it finishes, both tools are ready to use. **PM2** keeps the backend application running in the background and automatically restarts it if it ever crashes or the server reboots. **Serve** is used to display the website part of the application to users on the network.

3. Firewall settings

Two "doors" (called ports) need to be opened on this server so that other computers in the office can reach the application. These should **only** be open to computers on the office network, not to the wider internet, since this system holds employee data.

Port number	What it's for
5000	The backend, meaning the part that handles logins, saving data, and processing requests.
80	The frontend, meaning the actual website/screens that employees see and click through.

To open these:

1. Open Windows Defender Firewall with Advanced Security (search for it in the Start menu).
2. Click "Inbound Rules" on the left, then "New Rule..." on the right.
3. Choose "Port", click Next.
4. Choose "TCP" and enter `80,5000` under "Specific local ports". Click Next.
5. Choose "Allow the connection". Click Next.
6. Leave all three profiles (Domain, Private, Public) checked, unless your IT policy says otherwise. Click Next.
7. Give it a name, for example "Fute Portal Ports". Click Finish.

Extra safety step (recommended): if possible, restrict this rule further so it only accepts connections from the office's own network range (`192.168.1.0/24`) rather than from anywhere. This can be set under the rule's "Scope" tab by adding that range under "Remote IP address".

4. What happens after this

Once Node.js and MongoDB are installed, the replica set is turned on, and the firewall ports are open, IT's part of the setup is complete. The development team will then:

- Copy the application's code onto this server
- Run a few commands to install the application's own dependencies
- Start the backend using PM2 and the website using serve

Those remaining steps are covered separately in `SELF_HOSTED_LOCAL_MIGRATION.md` / `.pdf` , which is meant for the developer doing that part, not for IT.